Homework sheet 2: Multiple Quantum Systems

Problem 1. Bipartite states, bra-ket notation and partial traces. If we want to write a vector $|v\rangle \in \mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$ as an element of $\mathbb{C}^{d_1 d_2}$, we order the product standard basis of $\mathbb{C}^{d_1} \otimes \mathbb{C}^{d_2}$ lexicographically. For example, under this ordering $|01\rangle \in \mathbb{C}^2 \otimes \mathbb{C}^2$ is

$$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \in \mathbb{C}^4.$$

(a) Let $|\psi_{AB}\rangle = \mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B = \mathbb{C}^2 \otimes \mathbb{C}^3$ be the vector given by

$$|\psi_{AB}\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

in the lexicographically ordered product basis. Write $|\psi_{AB}\rangle$ in bra-ket notation in the product basis $|ab\rangle$ for $\mathcal{H}_{AB}$.

(b) Let $|\psi_{A'B'}\rangle = \mathcal{H}_{A'B'} = \mathcal{H}_{A'} \otimes \mathcal{H}_{B'} = \mathbb{C}^3 \otimes \mathbb{C}^2$ be the vector given by

$$|\psi_{A'B'}\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

in the lexicographically ordered product basis. Write $|\psi_{A'B'}\rangle$ in bra-ket notation in the *computational* for $\mathcal{H}_{A'B'}$.

(c) Let $V_{A'B' \to AB}$ denote the isometry $\mathcal{H}_{A'B'} \to \mathcal{H}_{AB}$ that swaps the two subsystems, i.e., $|xy\rangle \mapsto |yx\rangle$ for $x \in \{0, 1, 2\}$ and $y \in |0, 1\rangle$. Write down $V_{A'B' \to AB}|\psi_{A'B'}\rangle \in \mathcal{H}_{AB}$ as a 6-dimensional vector as well as in bra-ket notation.

(d) Let $\rho_{AB} = |\psi_{AB}\rangle\langle\psi_{AB}|$ and $\sigma_{A'B'} = |\psi_{A'B'}\rangle\langle\psi_{A'B'}|$. Compute the reduced density matrices ρ_A, ρ_B, $\sigma_{A'}$ and $\sigma_{B'}$.

Problem 2. Product measurement. For measurements $\mu_A : \Omega_1 \to \mathrm{PSD}(A)$ and $\mu_B : \Omega_2 \to \mathrm{PSD}(B)$ on quantum systems A and B, the *product measurement* $\mu_A \otimes \mu_B$ is defined by the formula

$$(\mu_A \otimes \mu_B)(x_1, x_2) = \mu_A(x_1) \otimes \mu_B(x_2),$$

for $x_1 \in \Omega_1$ and $x_2 \in \Omega_2$.

In the subproblems (c)–(e) assume that Alice and Bob share a maximally entangled state $|\Phi^+_{AB}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

(a) Show that the formula defines a measurement.

(b) Prove that if we measure any state $\rho_{AB} \in S(AB)$ using the product measurement, then the marginal probability of Alice's outcome $x_1 \in \Omega_1$ is the same as if Bob did not make any measurement at all.

(c) Suppose that Alice and Bob measure in the standard basis. What is the probability distribution of the outcomes?

(d) Suppose that Alice and Bob measure in the X-basis $|+\rangle$ and $|-\rangle$. What is the probability distribution of the outcomes?

(e) In both cases, what does the marginal distribution of the outcomes look like for Alice and Bob? Relate this to their reduced states. Note that while the measurement outcomes are correlated, performing local measurements on a maximally engangled state does not allow them to communicate information.

Problem 3. Purity of quantum states. The *purity* of a quantum state ρ is defined as $P(\rho) = \mathrm{tr}[\rho^2]$.

(a) For $\rho \in S(A)$, $\dim(\mathcal{H}_A) = d$, prove

$$\frac{1}{d} \leq P(\rho) \leq 1.$$

When is equality achived?

(b) Let $\rho_{AB} \in S(AB)$ be a pure state with marginal states ρ_A and ρ_B. Show that $P(\rho_A) = P(\rho_B)$.